IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART II: MEng, BEng and ACGI

MATHEMATICS 2A (E-STREAM AND I-STREAM)

Thursday, 21 May 10:00 am

Time allowed: 1:30 hours

There are SIX questions on this paper.

Answer ALL questions

NO CALCULATORS ALLOWED. Table of Laplace transforms included

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : O. Sydoruk

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Table of Laplace transforms

$f(t)$	$\mathcal{L}\{f(t)\} \equiv F(s)$
A	$\frac{A}{s}\,, \qquad \mathrm{Re}(s) > 0$
e^{at}	$\frac{1}{s-a}\,, \qquad \mathrm{Re}(s) > a$
$t^n, \qquad n = 1, 2 \ldots$	$\frac{n!}{s^{n+1}}\,, \qquad \mathrm{Re}(s) > 0$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}\,, \qquad \mathrm{Re}(s) > 0$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}\,, \qquad \mathrm{Re}(s) > 0$
$e^{at} f(t)$	$F(s-a)$
$t^n f(t)$	$(-1)^n \frac{d^n F}{ds^n}$
$\frac{df}{dt}$	$sF(s) - f(0)$
$\frac{d^2 f}{dt^2}$	$s^2 F(s) - sf(0) - \frac{df}{dt}(0)$
$H(t-a)$	$\frac{e^{-as}}{s}$
$\delta(t-a)$	$e^{-as}\,, \qquad a > 0$
$f(t-a)H(t-a)$	$e^{-as}F(s)$
$\int_0^t f(u)g(t-u)\ du$	$F(s)G(s)$

EE2-08A MATHEMATICS

1. Determine if the function $u(x,y) = xe^x \cos y - ye^x \sin y$ is harmonic, and if so, find its harmonic conjugate $v(x,y)$. [10]

2. Evaluate the complex integral

$$\oint_C \frac{2+3\sin(\pi z)}{z(z-1)^2}\, dz,$$

where C is the boundary of a square with vertices given by $\pm 3 \pm 3j$. [8]

3. Express $\cos\theta$ in terms of z, where $z = e^{j\theta}$ is on the unit circle, and rewrite the real integral

$$I = \int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos\theta}\, d\theta$$

as a complex integral around the unit circle. Hence, or otherwise, show that

$$I = \frac{\pi}{12}.$$ [12]

4. Let $w(z) = \sqrt{2}e^{j\pi/4}z + (1-2j)$, and the region R be bounded by the lines $x = 0, y = 0, x = 2$, and $y = 1$. Determine the region R' of the $w-$plane into which R is mapped under the transformation given by $w(z)$. Sketch both regions R and R'. [8]

5. Obtain, from first principles, the Laplace Transform of the periodic function given by

$$f(t) = \begin{cases} t, & 0 \le t < 1 \\ f(t-1), & t \ge 1\,. \end{cases}$$ [6]

6. Use Laplace Transforms to solve the ODE $\dfrac{d^2y}{dt^2} + y = t$, where $y(0) = 0$ and $y'(0) = 2$. [6]

Recall that the residue of a complex function $F(z)$ at a pole $z = a$ of multiplicity m is given by the expression

$$\lim_{z\to a}\left\{ \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m F(z)\right]\right\}.$$